

Representations of Numbers in Various Bases

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1 Introduction

Any fraction $\frac{1}{n}$ can be represented in a base b as a repeating or finite decimal. When all prime factors of n are also factors of b , $\frac{1}{n}$ will have a finite decimal expansion. When some of the factors of n are relatively prime to b , $\frac{1}{n}$ will repeat but may have terminating digits at the beginning. Purely periodic decimals occur when $(n, b) = 1$. The numerator need not be 1, but $\frac{1}{n}$ is the most fundamental case. In this article, I will focus on reciprocals of primes to ensure purely periodic decimals.

2 Numerical Data

To begin, I will perform a test where I take a fraction $\frac{1}{p}$ where p is a prime number, and find the period of the repeating digits in its decimal representation. For example, $\frac{1}{11} = 0.090909\dots$ so the period is 2 and the repeating digits are 0 and 9.

p	Period	Repeating Digits
3	1	3
7	6	142857
11	2	09
13	6	076923
17	16	0588235294117647
19	18	052631578947368421
23	22	0434782608695652173913
29	28	0344827586206896551724137931
31	15	032258064516129

37	3	027
41	5	02439
43	21	023255813953488372093
47	46	02127659574468085106382978723404...
53	13	0188679245283
59	58	01694915254237288135593220338983...
61	60	01639344262295081967213114754098...

Define a **reptend prime** as a prime with a cyclic decimal expansion of period $p - 1$. We can notice that the first few reptend primes are 7, 17, 19, 23, 29, 47, 59, 61, 97, 109, 113, 131, 149, 167, 179, 181, 193, ...

3 Period Lengths

Consider the repeating decimal of $\frac{1}{p}$ in base b .

3.0.1 Claim

The period length of $\frac{1}{p}$ is the order of b in U_p .

3.0.2 Proof

First, use the long division algorithm to divide 1 by p in base b . Start by writing 1 as 1.000.... In each step, one must “bring down” a zero and subtract a multiple of p ’s equivalent mod p when dividing by a number b , so long division effectively exponentiates $b \bmod p$. The decimal repeats when the remainder mod p is 1, which occurs when $b^k = 1 \pmod{p}$. Therefore, the period length of $\frac{1}{p}$ is the order of $b \pmod{p}$.

3.1 Lemma

The decimal expansion of $\frac{1}{p}$, where p is a prime number that is not 2 or 5, is purely recurring because the greatest power of 2 and 5 that divides p is 0. More generally, consider a fraction where $\frac{m}{n}$, where m and n are integers and $(m, n) = 1$.

3.1.1 Claim

The number of non-recurring digits is (a, b) , where a is the maximum power of 2 that divides mn and b is the maximum power of 5 that divides mn .

3.1.2 Proof

For any n , we have $n = 2^a \cdot 5^b \cdot m$, where $(m, 10) = 1$ and $m \geq 1$. Now, take the infinite series $\sum_{i \geq 1} a \cdot 10^i \bmod n$. By the Pigeonhole Principle, there must be at least two terms that are congruent mod n . We can set the first such pair of terms to be $a \cdot 10^k$ and $a \cdot 10^{k+l}$, which yields

$$10^k \equiv 10^{k+l} \pmod{n}$$

because the GCD of a and n is 1, as $\frac{a}{n}$ is the simplified fraction. We also have that $(10, m) = 1$, so $1 = 10^l \pmod{n} \Rightarrow 1 = 10^l \pmod{m}$ because $m|n$. Thus, we have

$$a \cdot 10^{k+j} = a \cdot 10^{k+l+m} \pmod{n}$$

for any arbitrary $j \geq 0$. Furthermore, the order l of $10 \pmod{m}$ is the length of the period. Both 2^α and 5^β have to divide a^k as they are relatively prime to $10^l - 1$. The smallest k is therefore the maximum of α and β , representing the length of the non-recurring part of the decimal expansion. Hence, for primes p other than 2 or 5, $\frac{1}{p}$ is purely recurring.

3.1.3 Example

Consider $\frac{m}{n}$ where $m = 3$ and $n = 7$. No factors of 2 or 5 go into 3 or 7, meaning that $a = 0$ and $b = 0$. Therefore, $(a, b) = 0$, and $\frac{3}{7} = 0.\overline{428571}$ has no non-recurring digits.

3.1.4 Example

Consider $\frac{m}{n}$ where $m = 5$ and $n = 8$. Since $5 = 5^1$ and $8 = 2^3$, we have $mn = (5^1)(2^3)$, meaning that $a = 3$ and $b = 1$. We have $(a, b) = 3$, and $\frac{5}{8} = 0.625$ has three non-recurring digits.

4 Period of a Multiple

4.1 Theorem

The period of $\frac{1}{pq}$ is the LCM of the periods of $\frac{1}{p}$ and $\frac{1}{q}$, where p and q are primes greater than or equal to 5.

4.2 Proof

For prime number p , let's set $m = \text{ord}_p(b)$. We have $p \mid -b^m - 1$, so we can also write $b^m = pD + 1$. Therefore,

$$\frac{1}{p}b^m = D + \frac{1}{p},$$

meaning that $\frac{1}{p}$ in base b can be written as a repeating decimal $0.\overline{D}$ with a period of m digits. When m is even, we get

$$pD = b^m - 1 \equiv (-1)^m - 1 = 0 \pmod{b+1} \Rightarrow pD = s(b+1).$$

If we also have $(p, b+1) = 1$, we can further show that $b+1 \mid D$.

Now, we know $1/p$ has period $m = \text{ord}_p(b)$ and $1/q$ has period $n = \text{ord}_q(b)$, and we also have $1/pq$ has period $k = \text{ord}_{pq}(b)$. Finally, we can prove that $k = [m, n]$:

$$\begin{aligned} m &= \text{ord}_p(b) \text{ and } p \mid b^k - 1 \Rightarrow m \mid k \\ n &\mid k \Rightarrow [m, n] \mid k \\ b^m - 1 &\mid b^{[m, n]} - 1 \Rightarrow p \mid b^{[m, n]} - 1 \\ q \mid b^{[m, n]} - 1 &\Rightarrow pq \mid b^{[m, n]} - 1 \Rightarrow k \mid [m, n] \Rightarrow k = [m, n]. \end{aligned}$$

4.3 Example

$\frac{1}{7}$ and $\frac{1}{11}$ have periods 6 and 2, so $\frac{1}{77}$ has period 6 because $[6,2] = 6$.

4.4 Example

$\frac{1}{13}$ and $\frac{1}{17}$ have periods 6 and 16, so $\frac{1}{77}$ has period 6 because $[6,16] = 48$.

5 Half Period Theorem

5.1 Theorem

For primes $p \neq 2, 5$, if $\text{ord}_p(b) = 2d$ for some $d \geq 1$, then

$$\frac{n}{p} = 0.\overline{a_1 a_2 \dots a_d a_{d+1} \dots a_{2d}},$$

where $a_k + a_{d+k} = b$ for $1 \leq k \leq d$.

5.2 Proof

We have the following:

$$\text{ord}_p(b) = 2d \Rightarrow b^{2d} = mp + 1 \Rightarrow \frac{n}{p} \times b^{2d} = mn + \frac{n}{p}$$

$$\frac{n}{p} = 0.\overline{D}, D = \frac{n}{p}(10^{2d} - 1) < b^{2d-1} \Rightarrow D = a_1 a_2 \dots a_d a_{d+1} \dots a_{2d}.$$

Let $A = a_1 a_2 a_3 \dots a_d$ and $B = a_{d+1} a_{d+2} a_{d+3} \dots a_{2d}$, so we have $D = b^d A + B$. Note that both A and B have length d . We also have $p \nmid b^d - 1$ because $2d$ is the order of b for $(\text{mod } p)$. Then,

$$pD = n(b^{2d} - 1) = n(b^d - 1)(b^d + 1) \Rightarrow p \mid n(b^d + 1) \text{ and } b^d + 1 \mid D$$

$$D = b^d A + B \equiv A + B \pmod{b^d - 1} \Rightarrow A + B = b^d - 1 \text{ because } A + B < 2(b^d - 1).$$

5.3 Example

The decimal expansion of $\frac{1}{7}$ is $7 \ 0.\overline{142857}$, and $142 + 857 = 999$.

5.4 Example

The decimal expansion of $\frac{1}{29}$ is $0.\overline{0344827586206896551724137931}$, and $142+857 = 999$. $03448275862068+96551724137931 = 99999999999999$.

6 Connections to Class Numbers

6.1 Conjecture

If $\frac{1}{p}$ has a cyclical permutation, then if the alternating sum of the digits in the decimal expansion is taken: $\overline{a_1 a_2 \dots a_d}$ then $\sum_{i=1}^d (-1)^i \cdot a_i$ is equal to 0 when the prime is of the form $4k + 1$ and $11h(-p)$ when the prime is of the form $4k + 3$. $h(-p)$ is equal to the class number of the primitive binary quadratic forms with discriminant $-p$.

6.2 Motivation

Below are a few values:

p	Sum	$h(-p)$
7	11	1
17	0	4
19	11	1
23	33	3
29	0	6
47	55	5
59	33	3
61	0	6
97	0	4

Fields looked into: Stark-Heegner Theorem, Imaginary Quadratic Fields, Ideals...

I explored different sums of $\frac{1}{p}$ which yielded the formula in Section 4, and eventually took a look at the alternating sums. I then ran a Python script for the numbers this sum yields for $\frac{1}{p}$. Some of the data displayed that some of the numbers with this sum as multiples of 11 are primes with period $p - 1$. This led to the splitting into cases of primes of the form $4k + 1$ and $4k + 3$. Primes of the form $4k + 1$ and $4k + 3$ had a sum of 11 multiplied by some integer.

6.3 Example

For the decimal expansion of $\frac{1}{23}$, this sum equals $33 = 3 \cdot 11$, where $3 = h(-23)$, the class number of -23 . The forms of the binary quadratics are $(a, b, c) = (1, 1, 6), (2, -1, 3), (2, 1, 3)$. 23 is additionally of the form $4k + 3$ where $k = 5$.

6.4 Example

For the decimal expansion of $\frac{1}{17}$, this sum equals 0. 17 is of the form $4k + 1$ where $k = 4$.

7 p -adic Integers and Decimal Expansions

7.1 Theorem

Let $f(x) \in \mathbb{Z}[x]$, $a, p \in \mathbb{Z}$, p prime, and suppose $f(a) \equiv 0 \pmod{p^n}$ and $f'(a) \not\equiv 0 \pmod{p}$. Then there is a unique $a_n \in \mathbb{Z}_{p^{n+1}}$ such that $f(a + a_n p^n) \equiv 0 \pmod{p^{n+1}}$.

7.2 Proof

We can set $m = \deg(f(x))$ and expand $f(x)$ as a Taylor Series at $x = a$ to get:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots + \frac{f^{(m)}(a)}{m!}(x - a)^m.$$

Next, evaluating at $x = a + a_n p^n$ yields

$$f(a + a_n p^n) = f(a) + f'(a)a_n p^n + \dots + \frac{f^{(m)}(a)}{m!}a_n^m p^{nm}.$$

Since we are working $(\text{mod } p^{n+1})$, we can reduce

$$f(a + a_n p^n) \equiv f(a) + f'(a) a_n p^n \pmod{p^{n+1}}$$

as all other terms have higher than p^{n+1} powers of p and have integer coefficients as you can express $\frac{1}{m!} f^{(m)}(x) = \binom{n}{m} x^{n-m}$.

We want to prove that there exists an a_n such that $f'(a) a_n p^n \equiv -f(a) \pmod{p^{n+1}}$. Because we have $p^n | f(a)$, we can use modular cancellation to get that

$$f'(a) a_n p^n \equiv -f(a) \pmod{p^{n+1}} \iff f'(a) a_n \equiv \frac{-f(a)}{p^n} \pmod{p}.$$

Using linear diophantine equations and the fact that $f'(a) \not\equiv 0 \pmod{p}$, there is a unique solution for $a_n \pmod{p}$. Thus, if we trace backwards, proving this means proving that $f(a + a_n p^n) \equiv 0 \pmod{p^{n+1}}$, thus completing the proof.

This can also be generalized to work over p -adic integers.

8 Polynomials

We can extend the ideas from considering fractions in base b to considering the fractions in various polynomial rings. However, first note the following remark.

8.0.1 Remark

A number in base b , such as $(a_1 a_2 a_3 \dots a_k)_b$ can simply be written as a polynomial in b . So, we can write

$$(a_1 a_2 a_3 \dots a_k)_b = a_1 b^{k-1} + a_2 b^{k-2} + a_3 b^{k-3} \dots + a_k.$$

Likewise, if we have a polynomial in b , call it $f(b)$, then we can view $f(b)$ as a number in base b by examining the coefficients of f .

For example, suppose we have the polynomials $\text{mod } p : \mathbb{Z}_p$. Now, consider the fractions $\frac{1}{x+u}$ of $\mathbb{Z}_p$, where u is a unit in $\mathbb{Z}_p$. Depending on the values of p and u , the polynomial expansion of $\frac{1}{x+u}$ in base p can either be infinite or finite.

8.0.2 Example

Suppose $p = 5$, so we are working in $\mathbb{Z}_5$. Consider the selection $u = 1$. Then we have that

$$\begin{aligned} \frac{1}{x+1} &= \frac{1}{1-(-x)} \\ &= 1 - x + x^2 - x^3 \dots \\ &= 1 + 4x + x^2 + 4x^3 \dots \text{ in } \mathbb{Z}_5, \end{aligned}$$

which is an infinite polynomial.

Note that this infinite polynomial only has coefficients 1, 4, which is not the entire set of units in $\mathbb{Z}_5$. Specifically, the polynomial does not have coefficients 2 or 3. This begs the question, when do the coefficients of the polynomial expansion of $\frac{1}{x+u}$ in $\mathbb{Z}_p$ cycle through all the elements of U_5 ? The following conjecture can thus be made:

8.1 Theorem (Generators Conjecture)

When written as a polynomial in $\mathbb{Z}_p$, the fraction $\frac{1}{x+u}$ will only cycle through all the elements of U_5 if $u = g$, where g is a generator mod p .

8.1.1 Example

As shown previously, since $u = 1$ is not a generator mod 5, the expansion of $\frac{1}{x+1}$ does not cycle through all coefficients. However, now suppose $u = 2$, which is a generator mod 5. This means that the polynomial expansion in $\mathbb{Z}_5$ is

$$\begin{aligned}\frac{1}{x+2} &= \frac{\frac{1}{2}}{\frac{x}{2}+1} \\ &= \frac{\frac{1}{2}}{\frac{x}{2}+1} \\ &= \frac{\frac{1}{2}}{1 - (-\frac{x}{2})} \\ &= \frac{3}{1 - (-3x)} \\ &= 3 - 9x + 27x^2 - 81x^3 + 243x^4 \dots \\ &= 3 + x + 2x^2 + 4x^3 \dots \text{ in } \mathbb{Z}_5\end{aligned}$$

which certainly hits all elements of U_5 .

9 Conclusion

When diving into the world of the representation of numbers, one can figure that there are many conjectures that are yet to be made. Although a lot of information can currently be found online, sometimes it is nice to relax and attempt to make conjectures and prove or salvage them on your own. The conjectures that I made on this topic range from periods to relations to calculus (Taylor function), and there are still many more phenomena that are to be desired in this field of mathematics.